

# JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

## BINOMIAL THEOREM

DIBY (DO IT BY YOURSELF)

By: Sachin Jakhar (Defender Mamu)





### # DIBY-01

The coefficient of  $x^{-6}$  in the expansion of  $\left(\frac{4x}{5} + \frac{5}{2x^2}\right)^9$ , is \_\_\_\_\_

[JEE Mains-2023]

[Ans. : 5040]

### # DIBY-02

The natural number  $m$ , for which the coefficient of  $x$  in the binomial expansion of

$\left(x^m + \frac{1}{x^2}\right)^{22}$  is 1540, is \_\_\_\_\_

[JEE Mains-2020]

[Ans. : 13]

### # DIBY-03

The sum of the real values of  $x$  for which the middle term in the binomial expansion of

$\left(\frac{x^3}{3} + \frac{3}{x}\right)^8$  equals 5670 is

[JEE Mains-2019]

(A) 0                      (B) 8                      (C) 6                      (D) 4

[Ans. : (A)]

### # DIBY-04

The ratio of the coefficient of the middle term in the expansion of  $(1 + x)^{20}$  and the sum of the coefficients of two middle terms in the expansion of  $(1 + x)^{19}$  is

[JEE Mains-2021]

[Ans. : 1]



### # DIBY-05

If the constant term, in binomial expansion of  $\left(2x^r + \frac{1}{x^2}\right)^{10}$  is 180, then  $r$  is equal to

[JEE Mains-2021]

[Ans. : ]

### # DIBY-06

If the term independent of  $x$  in the expansion of  $\left(\sqrt{a}x^2 + \frac{1}{2x^3}\right)^{10}$  is 105, then  $a^2$  is equal to

[JEE Mains-2024]

(A) 2

(B) 4

(C) 6

(D) 9

[Ans. : (B)]

### # DIBY-07

If the constant term in the expansion of  $\left(\frac{\sqrt[5]{3}}{x} + \frac{2x}{\sqrt[3]{5}}\right)^{12}$ ,  $x \neq 0$  is  $\alpha \times 2^8 \times \sqrt[5]{3}$ , then  $25\alpha$  is equal to

[JEE Mains-2024]

[Ans. : 693]



### # DIBY-08

If the term without  $x$  in the expansion of  $\left(x^{2/3} + \frac{\alpha}{x^3}\right)^{22}$  is 7315, then  $|\alpha|$  is equal to

[JEE Mains-2023]

[Ans. : 1]

### # DIBY-09

If the coefficient of  $x^4$ ,  $x^5$  and  $x^6$  in the expansion of  $(1 + x)^n$  are the arithmetic progression then the maximum value of  $n$  is

[JEE Mains-2024]

- (A) 7                      (B) 21                      (C) 28                      (D) 14

[Ans. : (D)]

### # DIBY-10

Let the coefficient of three consecutive terms in the binomial expansion of  $(1 + 2x)^n$  be in the ratio  $2 : 5 : 8$ . Then the coefficient of the term, which is in the middle of these three terms, is

[JEE Mains-2023]

[Ans. : 1120]



### # DIBY-11

If the coefficient of  $x^7$  in  $\left(x^2 + \frac{1}{bx}\right)^{11}$  and  $x^{-7}$  in  $\left(x - \frac{1}{bx^2}\right)^{11}$  ( $b \neq 0$ ), are equal, then the value of  $b$  is equal to **[JEE Mains-2021]**

- (A) 2                      (B) -1                      (C) 1                      (D) -2

**[Ans.: (c)]**

### # DIBY-12

If the co-efficient of  $x^9$  in  $\left(\alpha x^3 + \frac{1}{\beta x}\right)^{11}$  and the co-efficient  $x^{-9}$  in  $\left(\alpha x - \frac{1}{\beta x^3}\right)^{11}$  are equal to, then  $(\alpha\beta)^2$  is equal to **[JEE Mains-2023]**

**[Ans.: 1]**



### # DIBY-13

If the coefficient of  $x^7$  in  $\left(ax - \frac{1}{bx^2}\right)^{13}$  and the coefficient of  $x^{-5}$  in  $\left(ax + \frac{1}{bx^2}\right)^{13}$  are equal, then  $a^4b^4$  is equal to **[JEE Mains-2023]**

- (A) 22                      (B) 33                      (C) 44                      (D) 11

**[Ans.: (A)]**

### # DIBY-14

If the second, third and fourth terms in the expansion of  $(x + y)^n$  are 135, 30 and  $10/3$ , respectively, then  $6(n^3 + x^2 + y)$  is equal to **[JEE Mains-2024]**

**[Ans.: 806]**

### # DIBY-15

The sum of the coefficient of  $x^{2/3}$  and  $x^{-2/5}$  in the binomial expansion of  $\left(x^{2/3} + \frac{1}{2}x^{-2/5}\right)^9$  is

**[JEE Mains-2024]**

- (A)  $21/4$                       (B)  $63/16$                       (C)  $19/4$                       (D)  $69/16$

**[Ans.: (A)]**

### # DIBY-16

If  ${}^nC_4$ ,  ${}^nC_5$  and  ${}^nC_6$  are in A.P., then  $n$  can be

- (A) 9 (B) 14 (C) 11 (D) 12

[JEE Mains-2019]

[Ans. : (B)]

### # DIBY-17

In the expansion  $\left(\frac{x}{\cos\theta} + \frac{1}{x\sin\theta}\right)^{16}$ , if  $l_1$  is the least value of the term independent of  $x$  when  $\frac{\pi}{8} \leq \theta \leq \frac{\pi}{4}$  and  $l_2$  is the least value of the term independent of  $x$  when  $\frac{\pi}{16} \leq \theta \leq \frac{\pi}{8}$ , then the ratio  $l_2 : l_1$  is equal to

- (A) 1:8 (B) 1:16 (C) 8:1 (D) 16:1

[JEE Mains-2020]

[Ans. : (D)]

### # DIBY-18

Let  $\alpha$  be the constant term in the binomial expansion of  $\left(\sqrt{x} - \frac{6}{x^{3/2}}\right)^n$ ,  $n \leq 15$ . If the sum of the coefficients of the remaining terms in the expansion is 649 and the coefficient of  $x^{-n}$  is  $\lambda\alpha$ , then  $\lambda$  is equal to

[JEE Mains-2023]

[Ans. : 36]

### # DIBY-19



The sum of the coefficient of  $x^{2/3}$  and  $x^{-2/5}$  in the binomial expansion of  $\left(x^{2/3} + \frac{1}{2}x^{-2/5}\right)^9$  is

[JEE Mains-2024]

- (A)  $\frac{21}{4}$  (B)  $\frac{69}{16}$  (C)  $\frac{63}{16}$  (D)  $\frac{19}{4}$

[Ans. : (A)]

### # DIBY-20

Let  $0 \leq r \leq n$ . If  ${}^{n+1}C_{r+1} : {}^nC_r : {}^{n-1}C_{r-1} = 55 : 35 : 21$ , then  $2n + 5r$  is equal to [JEE Mains-2024]

- (A) 60 (B) 62 (C) 50 (D) 55

[Ans. : (c)]

### # DIBY-21

For some  $n \neq 10$ , let the coefficients of the 5<sup>th</sup>, 6<sup>th</sup> and 7<sup>th</sup> terms in the binomial expansion of  $(1+x)^{n+4}$  be in A.P. Then the largest coefficient in expansion of  $(1+x)^{n+4}$  is

[JEE Mains-2025]

- (A) 35 (B) 20 (C) 10 (D) 70

[Ans. : (A)]

### # DIBY-22

A ratio of the 5<sup>th</sup> term from the beginning to the 5<sup>th</sup> term from the end in the binomial

expansion of  $\left(2^{1/3} + \frac{1}{2(3)^{1/3}}\right)^{10}$  is

[JEE Mains-2019]

- (A)  $1:2(6)^{1/3}$  (B)  $1:4(16)^{1/3}$  (C)  $4(36)^{1/3}:1$  (D)  $2(36)^{1/3}:1$

[Ans.: (C)]

### # DIBY-23

If  $\alpha$  and  $\beta$  be the coefficients of  $x^4$  and  $x^2$  respectively in the expansion of

$(x + \sqrt{x^2 - 1})^6 + (x - \sqrt{x^2 - 1})^6$ , then

[JEE Mains-2020]

- (A)  $\alpha + \beta = -30$  (B)  $\alpha - \beta = 60$  (C)  $\alpha - \beta = -132$  (D)  $\alpha + \beta = 60$

[Ans.: (C)]

### # DIBY-24

If the maximum value of the term independent of  $t$  in the expansion of

$\left(t^2 x^{1/5} + \frac{(1-x)^{1/10}}{t}\right)^{15}$ ,  $x \geq 0$  is  $K$ , then  $8K$  is equal to

[JEE Mains-2022]

[Ans.: 6006]



### # DIBY-25

Let the coefficients of  $x^{-1}$  and  $x^{-3}$  in the expansion of  $\left(2x^{1/5} - \frac{1}{x^{1/5}}\right)^{15}$   $x > 0$  be  $m$  and  $n$  respectively. If  $r$  is a positive integer such that  $mn^2 = {}^{15}C_r \cdot 2^r$ , then the value of  $r$  is equal to  
[JEE Mains-2022]  
[Ans. : 5]

### # DIBY-26

If  $a, b$  and  $c$  are the greatest values of  ${}^{19}C_p$ ,  ${}^{20}C_q$  and  ${}^{21}C_r$ , respectively, then  
[JEE Mains-2020]  
(A)  $\frac{a}{11} = \frac{b}{11} = \frac{c}{42}$  (B)  $\frac{a}{11} = \frac{b}{11} = \frac{c}{42}$  (C)  $\frac{a}{11} = \frac{b}{11} = \frac{c}{42}$  (D)  $\frac{a}{11} = \frac{b}{11} = \frac{c}{42}$   
[Ans. : (c)]

### # DIBY-27

Let the ratio of the fifth term from the beginning to the fifth term from the end in the binomial expansion of  $\left(\sqrt[4]{2} + \frac{1}{\sqrt[4]{3}}\right)^n$ , in the increasing powers of  $\frac{1}{\sqrt[4]{3}}$  be  $\sqrt[4]{6} : 1$ . If the sixth term from the beginning is  $\frac{\alpha}{\sqrt[4]{3}}$ , then  $\alpha$  is equal to  
[JEE Mains-2022]  
[Ans. : 84]

## NUMBER OF RATIONAL OR IRRATIONAL TERMS



### # DIBY-28

If  $n$  is the number of irrational terms in the expansion of  $(3^{1/4} + 5^{1/8})^{60}$ , then  $(n - 1)$  is divisible by **[JEE Mains-2021]**

- (A) 26                      (B) 30                      (C) 8                      (D) 7

**[Ans. : (A)]**

### # DIBY-29

The sum of all rational terms in the expansion of  $(2^{1/5} + 5^{1/3})^{15}$  is equal to **[JEE Mains-2024]**

- (A) 3133                      (B) 633                      (C) 931                      (D) 6131

**[Ans. : (A)]**

### # DIBY-30

The Least value of  $n$  for which the number of integral terms in the Binomial expansion of  $(\sqrt[3]{7} + \sqrt[12]{11})^n$  is 183, is **[JEE Mains-2025]**

- (A) 2148                      (B) 2184                      (C) 2196                      (D) 2172

**[Ans. : (B)]**

## NUMBER OF RATIONAL OR IRRATIONAL TERMS



# DIBY-31

The number of integral terms in the expansion of  $(5^{1/2} + 7^{1/8})^{1016}$  is **[JEE Mains-2025]**

- (A) 127                      (B) 130                      (C) 129                      (D) 128

**[Ans. : (D)]**

# DIBY-32

The total number of irrational terms in the binomial expansion of  $(7^{1/5} - 3^{1/10})^{60}$  is

**[JEE Mains-2019]**

- (A) 55                      (B) 49                      (C) 48                      (D) 54

**[Ans. : 2019]**

# DIBY-33

The number of rational terms in the binomial expansion of  $(4^{1/4} + 5^{1/6})^{120}$  is

**[JEE Mains-2021]**

**[Ans. : 21]**

## NUMBER OF RATIONAL OR IRRATIONAL TERMS



### # DIBY-34

The sum of all those terms which are rational numbers in the expansion of

$$(2^{1/3} + 3^{1/4})^{12} \text{ is}$$

[JEE Mains-2021]

- (A) 89                      (B) 35                      (C) 27                      (D) 43

[Ans. : (D)]

### # DIBY-35

Number of integral terms in the expansion of  $\{7^{(1/2)} + 11^{(1/6)}\}^{824}$  is equal to

[JEE Mains-2024]

[Ans. : 138]

### # DIBY-36

The sum of all rational terms in the expansion of  $(2 + \sqrt{3})^8$  is

[JEE Mains-2025]

- (A) 16923                      (B) 3763                      (C) 33845                      (D) 18817

[Ans. (D)]

## FINDING COEFFICIENTS



### # DIBY-37

The coefficient of  $x^{101}$  in the expression

$(5 + x)^{500} + x(5 + x)^{499} + x^2(5 + x)^{498} + \dots + x^{500}, x > 0$ , is

[JEE Mains-2022]

- (A)  ${}^{501}C_{101}(5)^{399}$  (B)  ${}^{501}C_{101}(5)^{400}$  (C)  ${}^{501}C_{100}(5)^{400}$  (D)  ${}^{500}C_{101}(5)^{309}$

[Ans. : (A)]

### # DIBY-38

The coefficient of  $x^{301}$  in  $(1 + x)^{500} + x(1 + x)^{499} + x^2(1 + x)^{498} + \dots + x^{500}$  is: [JEE Mains-2023]

- (A)  ${}^{501}C_{302}$  (B)  ${}^{500}C_{301}$  (C)  ${}^{500}C_{300}$  (D)  ${}^{501}C_{200}$

[Ans. : (D)]

### # DIBY-39

The coefficient of  $x^{70}$  in  $x^2(1 + x)^{98} + x^3(1 + x)^{97} + x^4(1 + x)^{96} + \dots + x^{54}(1 + x)^{46}$  is

${}^{99}C_p - {}^{46}C_q$ . Then a possible value to  $p + q$  is:

[JEE Mains-2024]

- (A) 55 (B) 61 (C) 68 (D) 83

[Ans. : (D)]

## FINDING COEFFICIENTS



### # DIBY-40

The coefficient of  $x^7$  in the expression  $(1+x)^{10} + x(1+x)^9 + x^2(1+x)^8 + \dots + x^{10}$  is:

[JEE Mains-2020]

(A) 210

(B) 330

(C) 420

(D) 120

[Ans.: (B)]

### # DIBY-41

The coefficient of  $x^{256}$  in the expansion of  $(1-x)^{101}(x^2+x+1)^{100}$  is:

[JEE Mains-2021]

(A)  $-{}^{100}C_{15}$

(B)  ${}^{100}C_{16}$

(C)  $-{}^{100}C_{16}$

(D)  ${}^{100}C_{15}$  [Ans.: (D)]

### # DIBY-42

The coefficient of  $x^{2012}$  in the expansion of  $(1-x)^{2008}(1+x+x^2)^{2007}$  is equal to

[JEE Mains-2024]

[Ans.: (0)]



## FINDING COEFFICIENTS

# DIBY-43

The coefficient of  $x^4$  in the expansion of  $(1 + x + x^2 + x^3)^6$  in powers of  $x$ , is

[JEE Mains-2020]

[Ans. : (120)]

# DIBY-44

The term independent of  $x$  in the expansion of  $(1 - x^2 + 3x^3) \left(\frac{5}{2}x^3 - \frac{1}{5x^2}\right)^{11}$ ,  $x \neq 0$  is:

[JEE Mains-2022]

(A)  $\frac{7}{40}$

(B)  $\frac{33}{200}$

(C)  $\frac{39}{200}$

(D)  $\frac{11}{50}$

[Ans. : (B)]

# DIBY-45

If  $\binom{40}{0} + \binom{41}{1} + \binom{42}{2} + \dots + \binom{60}{20} = \frac{m}{n} \binom{60}{20}$ ,  $m$  and  $n$  are coprime, then  $m + n$  is equal to

[JEE Mains-2022]

[Ans. : 102]

# DIBY-46

If  $1^2 \cdot \binom{15}{1} + 2^2 \cdot \binom{15}{2} + 3^2 \cdot \binom{15}{3} + \dots + 15^2 \cdot \binom{15}{15} = 2^m \cdot 3^n \cdot 5^k$ , where  $m, n, k \in \mathbb{N}$ , then  $m + n + k$  is equal to

(A) 19

(B) 21

(C) 18

(D) 20

[Ans. : (A)]

# DIBY-47

The sum of series  $2 \times 1 \times {}^{20}C_4 - 3 \times 2 \times {}^{20}C_5 + 4 \times 3 \times {}^{20}C_6 - 5 \times 4 \times {}^{20}C_7 + \dots + 18 \times 17 \times {}^{20}C_{20}$  is equal to \_\_\_\_

[JEE Mains-2025]

[Ans. : 37]

# DIBY-48

If  $\sum_{r=0}^5 \frac{{}^{11}C_{2r+1}}{2r+2} = \frac{m}{n}$ ,  $\gcd(m, n) = 1$ , then  $m - n$  is equal to

[JEE Mains-2025]

[Ans. : 2035]

# DIBY-49

If  $\sum_{r=1}^{30} \frac{r^2 ({}^{30}C_r)^2}{{}^{30}C_{r-1}} = \alpha \times 2^{29}$ , then  $\alpha$  is equal to

[JEE Mains-2025]

[Ans. : 465]

# DIBY-50

Let  $\alpha = \sum_{r=0}^n (4r^2 + 2r + 1)^n C_r$  and  $\beta = \left( \sum_{r=0}^n \frac{{}^nC_r}{r+1} \right) + \frac{1}{n+1}$ . If  $140 < \frac{2\alpha}{\beta} < 281$ , then the value of  $n$  is

[JEE Mains-2024]

[Ans. : 5]

## SERIES

### # DIBY-51

The value of  $\frac{1}{1!50!} + \frac{1}{3!48!} + \frac{1}{5!46!} + \dots + \frac{1}{49!22!} + \frac{1}{51!1!}$  is

- (A)  $\frac{2^{50}}{50!}$  (B)  $\frac{2^{50}}{51!}$  (C)  $\frac{2^{51}}{51!}$  (D)  $\frac{2^{51}}{50!}$

[JEE Mains-2023]

[Ans. : (B)]

### # DIBY-52

If  $\sum_{r=0}^{10} \left( \frac{10^{r+1}-1}{10^r} \right) \cdot {}^{11}C_{r+1} = \frac{\alpha^{11}-11^{11}}{10^{10}}$ , then  $\alpha$  is equal to:

- (A) 15 (B) 11 (C) 24 (D) 20

[JEE Mains-2025]

[Ans. : (A)]

### # DIBY-53

If  $\sum_{r=1}^9 \left( \frac{r+3}{2^r} \right) \cdot {}^9C_r = \alpha \left( \frac{3}{2} \right)^9 - \beta$ ,  $\alpha, \beta \in N$ , then  $(\alpha + \beta)^2$  is equal to

- (A) 27 (B) 9 (C) 81 (D) 18

[JEE Mains-2025]

[Ans. : (C)]

## SERIES



### # DIBY-54

If  $({}^{30}C_1)^2 + 2({}^{30}C_2)^2 + 3({}^{30}C_3)^2 + \dots + 30({}^{30}C_{30})^2 = \frac{\alpha \cdot 60!}{(30!)^2}$ , then  $\alpha$  is equal to **[JEE Mains-2023]**

(A) 30

(B) 60

(C) 15

(D) 10

**[Ans. : (C)]**

### # DIBY-55

If  $a_r$  is the coefficient of  $x^{10-r}$  in the Binomial expansion of  $(1+x)^{10}$ , then  $\sum_{r=1}^{10} r^3 \left( \frac{a_r}{a_{r-1}} \right)^2$  is equal to **[JEE Mains-2023]**

(A) 4895

(B) 1210

(C) 5445

(D) 3025

**[Ans. : (B)]**

### # DIBY-56

If  $1 + (2 + {}^{49}C_1 + {}^{49}C_2 + \dots + {}^{49}C_{49})({}^{50}C_2 + {}^{50}C_4 + \dots + {}^{50}C_{50})$  is equal to  $2^n \cdot m$ , where  $m$  is odd, then  $n + m$  is equal to \_\_\_\_ **[JEE Mains-2022]**

**[Ans. : 99]**

## SERIES



### # DIBY-57

If  $n \geq 2$  is a positive integer, then the sum of the series  ${}^{n+1}C_2 + 2({}^2C_2 + {}^3C_2 + {}^4C_2 + \dots + {}^nC_2)$  is:  
[JEE Mains-2021]

- (A)  $\frac{n(n+1)^2(n+2)}{12}$  (B)  $\frac{n(n+1)(2n+1)}{6}$  (C)  $\frac{n(n-1)(2n+1)}{6}$  (D)  $\frac{n(2n+1)(3n+1)}{6}$

[Ans.: (B)]

### # DIBY-58

Let  $\binom{n}{k}$  denote  ${}^nC_k$  and  $\begin{bmatrix} n \\ k \end{bmatrix} = \begin{cases} \binom{n}{k}, & \text{if } 0 \leq k \leq n \\ 0, & \text{otherwise} \end{cases}$

If  $A_k = \sum_{i=0}^9 \binom{9}{i} \begin{bmatrix} 12 \\ 12 - k + i \end{bmatrix} + \sum_{i=0}^8 \binom{8}{i} \begin{bmatrix} 13 \\ 13 - k + i \end{bmatrix}$  and  $A_4 - A_3 = 190p$ , then  $p$  is equal to \_\_\_\_

[JEE Mains-2021]

[Ans.: 49]

### # DIBY-59

If  ${}^1P_1 + 2 \cdot {}^2P_2 + 3 \cdot {}^3P_3 + \dots + 15 \cdot {}^{15}P_{15} = {}^qP_r - s$ ,  $0 \leq s \leq 1$ , then  ${}^{q+s}C_{r-s}$  is equal to

[JEE Mains-2021]

[Ans.: 136]

## SERIES

### # DIBY-60

If  $\sum_{k=1}^{31} ({}^{31}C_k)({}^{31}C_{k-1}) - \sum_{k=1}^{30} ({}^{30}C_k)({}^{30}C_{k-1}) = \frac{\alpha(60!)}{(30!)(31!)}$ , where  $\alpha \in R$ , then the value of  $16\alpha$  is equal to **[JEE Mains-2022]**

- (A) 1411                      (B) 1320                      (C) 1615                      (D) 1855

**[Ans. : (A)]**

### # DIBY-61

If  $\sum_{i=1}^{20} \left( \frac{{}^{20}C_{i-1}}{{}^{20}C_i + {}^{20}C_{i-1}} \right)^3 = \frac{k}{21}$ , then  $k$  equals: **[JEE Mains-2019]**

- (A) 400                      (B) 50                      (C) 200                      (D) 100

**[Ans. : (D)]**

# DIBY-62

2.  ${}^{20}C_0 + 5. {}^{20}C_1 + 8. {}^{20}C_2 + 11. {}^{20}C_3 + \dots + 62. {}^{20}C_{20}$  is equal to

[JEE Mains-2019]

- (A)  $2^{23}$  (B)  $2^{26}$  (C)  $2^{24}$  (D)  $2^{25}$

[Ans. : (D)]

# DIBY-63

Let  $S_n = 1 + q + q^2 + \dots + q^n$  and  $T_n = 1 + \left(\frac{q+1}{2}\right) + \left(\frac{q+1}{2}\right)^2 + \dots + \left(\frac{q+1}{2}\right)^n$  where  $q$  is a real number and  $q \neq 1$ . If  ${}^{101}C_1 + {}^{101}C_2 \cdot S_1 + \dots + {}^{101}C_{101} \cdot S_{100} = \alpha T_{100}$  then  $\alpha$  is equal to: [JEE Mains-2019]

- (A)  $2^{99}$  (B) 202 (C) 200 (D)  $2^{100}$

[Ans. : (D)]

# DIBY-64

Let  $C_r$  denote the binomial coefficient of  $x^r$  in the expansion of  $(1 + x)^{10}$ . If  $\alpha, \beta \in \mathbb{R}$ .

1.1  $C_1 + 3.2 C_2 + 5.3 C_3 + \dots$  up to 10 terms  $= \frac{\alpha \times 2^{11}}{2^{\beta}-1} \left( C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots \text{upto 10 terms} \right)$ , then the value of  $\alpha + \beta$  is equal to

[JEE Mains-2022]

[Ans. : 286]

# DIBY-65

If  $\frac{1}{n+1} {}^nC_n + \frac{1}{n} {}^nC_{n-1} + \dots + \frac{1}{2} {}^nC_1 + {}^nC_0 = \frac{1023}{10}$ , then  $n$  is equal to

(A) 9

(B) 8

(C) 7

(D) 6

[JEE Mains-2023]

[Ans. : (A)]

### # DIBY-66

If for some  $m, n$ ;  ${}^6C_m + 2({}^6C_{m+1}) + {}^6C_{m+2} > {}^8C_3$  and  ${}^{n-1}P_3 : {}^nP_4 = 1 : 8$ , then  ${}^nP_{m+1} + {}^{n+1}C_m$  is  
[JEE Mains-2024]

- (A) 380 (B) 376 (C) 384 (D) 372

[Ans.: (D)]

### # DIBY-67

The value of  $\binom{30}{0}\binom{30}{10} - \binom{30}{1}\binom{30}{11} + \binom{30}{2}\binom{30}{12} \dots + \binom{30}{20}\binom{30}{30}$  is where  $\binom{n}{r} = {}^nC_r$  [IIT-JEE-2005]

- (A)  $\binom{30}{10}$  (B)  $\binom{30}{15}$  (C)  $\binom{60}{30}$  (D)  $\binom{31}{10}$

[Ans.: (A)]

### # DIBY-68

If  $a_n = \sum_{r=0}^n \frac{1}{{}^nC_r}$ , then value of  $\sum_{r=0}^n \frac{r}{{}^nC_r} =$

- (A)  $2a_n$  (B)  $na_n$  (C)  $\frac{2}{a_n}$  (D)  $\frac{na_n}{2}$

[IIT-JEE]

[Ans.: (D)]

### # DIBY-69

**Prove that**  ${}^{100}C_0 {}^{100}C_2 + {}^{100}C_2 {}^{100}C_4 + {}^{100}C_4 {}^{100}C_6 + \dots + {}^{100}C_{98} {}^{100}C_{100} = \frac{1}{2} [{}^{200}C_{98} - {}^{100}C_{49}]$ .

### # DIBY-70

**If**  $1 {}^{100}C_6 + 4 {}^{100}C_7 + 6 {}^{100}C_8 + 4 {}^{100}C_9 + {}^{100}C_{10}$  **has the value of**  ${}^xC_y$ , **then the value of**  $(x + y)$  **can be**

- (A) 114                      (B) 115                      (C) 198                      (D) 199

**[Ans. : (A), (C)]**

### # DIBY-71

**$\left(\frac{{}^nC_0 + {}^nC_1}{{}^nC_0}\right) \left(\frac{{}^nC_1 + {}^nC_2}{{}^nC_1}\right) \dots \left(\frac{{}^nC_{n-1} + {}^nC_n}{{}^nC_{n-1}}\right) = \frac{(15)^{14}}{14!}$ , find value of**  $n$ .

**[Ans. : 14]**

### # DIBY-72

$$\sum_{r=1}^n \sum_{p=0}^{r-1} {}^nC_r \cdot {}^rC_p \cdot 2^p$$

[Ans. :  $4^n - 3^n$ ]

### # DIBY-73

Prove that  $C_0 - 2^2 \cdot C_1 + 3^2 \cdot C_2 - \dots + (-1)^n (n+1)^2 C_n = 0, n > 2$ , where  $C_r = {}^nC_r$

[JEE 1989]

### # DIBY-74

If the value of  ${}^{55}C_5 - {}^5C_1 \cdot {}^{44}C_5 + {}^5C_2 \cdot {}^{33}C_5 - {}^5C_3 \cdot {}^{22}C_5 + {}^5C_4 \cdot {}^{11}C_5$  is  $a^b$  where  $a$  and  $b$  are both prime numbers, then find  $(a - b) =$

[Ans. : 6]

# DIBY-75

Let  $S_1 = \sum_{j=1}^{10} j(j-1) {}^{10}C_j$ ,  $S_2 = \sum_{j=1}^{10} j {}^{10}C_j$  and  $S_3 = \sum_{j=1}^{10} j^2 {}^{10}C_j$

Statement-1:  $S_3 = 55 \times 2^9$

Statement-2:  $S_1 = 90 \times 2^8$  and  $S_2 = 10 \times 2^8$

[AIEEE-2010]

- (A) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1
- (B) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1
- (C) Statement-1 is true, Statement-2 is false
- (D) Statement-1 is false, Statement-2 is true

[Ans.: (c)]

### # DIBY-76

**Prove that**  $\frac{1^2}{3} {}^nC_1 + \frac{1^2+2^2}{5} {}^nC_2 + \frac{1^2+2^2+3^2}{7} {}^nC_3 + \dots + \frac{1^2+2^2+\dots+n^2}{2n+1} {}^nC_n = \frac{n(n+3)}{6} 2^{n-2}$

### # DIBY-77

**If**  $C_r = {}^nC_r$ , **then prove that**  $(C_0 + C_1)(C_1 + C_2) \cdots (C_{n-1} + C_n) = (C_1 C_2 \cdots C_{n-1} C_n)(n+1)^n/n!$ .

### # DIBY-78

**Find the coefficient of**  $a^3 b^4 c^5$  **in the expansion of**  $(bc + ca + ab)^6$ .

**[Ans. : (60)]**



# DIBY-79

The coefficient of  $x^4$  in the expansion of  $(1 + x + x^2)^{10}$  is

[JEE Mains-2020]

[Ans. : 615]

# DIBY-80

The sum of all rational terms in the expansion of  $(1 + 2^{1/3} + 3^{1/2})^6$  is equal to \_\_\_\_

[JEE Mains-2025]

[Ans. : 612]

# DIBY-81

If the constant term in the expansion of  $(3x^3 - 2x^2 + \frac{5}{x^5})^{10}$  is  $2^k \cdot \ell$ , where  $\ell$  is an odd integer, then the value of  $k$  is equal to:

[JEE Mains-2022]

(A) 6

(B) 7

(C) 8

(D) 9

[Ans. : (D)]

### # DIBY-82

Prove that:  $\sum_{s=1}^n \sum_{r=0}^s {}^nC_s \cdot {}^sC_r = 3^n - 1$

### # DIBY-83

Find the value of  $\sum_{0 \leq i < j \leq n} ({}^nC_i + {}^nC_j)$

[Ans. :  $n2^n$ ]

### # DIBY-84

Find the value of  $\sum_{0 \leq i < j \leq n} (i + j)({}^nC_i + {}^nC_j)$

[Ans. :  $n^2 2^n$ ]

## FINDING REMAINDER



### # DIBY-85

Find remainder when  $2021^{2023}$  is divided by 7.

[JEE Mains-2022]

[Ans. : 5]

### # DIBY-86

The remainder, when  $7^{103}$  is divided by 17, is [JEE Mains-2023]

[Ans. : 12]

### # DIBY-87

Find the remainder when

(i)  $7^{103}$  is divided by 25.

[Ans. : 18]

(ii)  $2345^{678} + 1234^{567}$  is divided by 4.

[Ans. : 1]

### # DIBY-88

Find remainder when  $2021^{2022} + 2022^{2021}$  is divided by 7.

[JEE Mains-2022]

[Ans. : 0]

## FINDING REMAINDER



### # DIBY-89

Fractional part of the number  $4^{2022}/15$  is equal to

[JEE Mains-2023]

- (A)  $8/15$  (B)  $4/15$  (C)  $14/15$  (D)  $1/15$

[Ans. : (D)]

### # DIBY-90

Let the number  $(22)^{2022} + (2022)^{22}$  leave the remainder  $\alpha$  when divided by 3 and  $\beta$  when divided by 7. Then  $(\alpha^2 + \beta^2)$  is equal to

[JEE Mains-2023]

- (A) 20 (B) 13 (C) 5 (D) 10

[Ans. : (C)]

### # DIBY-91

Remainder when  $2^{100} + 3^{100} + 4^{100} + 5^{100}$  is divided by 7 is

[Ans. : 5]

### # DIBY-92

Use binomial theorem to prove that  $11^n - 10n - 1$  is divisible by 100  $\forall n \in \mathbb{N}$ .

## FINDING REMAINDER



### # BUMPER DIBY-93

- (i) When  $5^{97}$  is divided by 52, the remainder is  
(A) 3 (B) 4 (C) 5 (D) 0
- (ii) When  $13^{99} - 19^{92}$  is divided by 162, the remainder is  
(A) 3 (B) 4 (C) 5 (D) 0
- (iii) When  $27^{10} + 7^{51}$  is divided by 10, the remainder is  
(A) 0 (B) 2 (C) 4 (D) 5
- (iv) When  $106^{85} - 85^{106}$  is divided by 7, the remainder is  
(A) 0 (B) 1 (C) 2 (D) 3
- (v) When  $53^{53} - 33^3$  is divided by 10, the remainder is  
(A) 0 (B) 2 (C) 4 (D) 6

[Ans. : (i) (C), (ii) (D), (iii) (B), (iv) (A), (v) (D)]

### # DIBY-94

If the fractional part of the number  $\frac{2^{403}}{15}$  is  $\frac{k}{15}$ , then  $k$  is equal to

[JEE Mains-2019]

- (A) 8                      (B) 4                      (C) 6                      (D) 14

[Ans.: (A)]

### # DIBY-95

The remainder left out when  $8^{2n} - (62)^{2n+1}$  is divided by 9 is

[AIEEE-2009]

- (A) 2                      (B) 7                      (C) 8                      (D) 0

[Ans.: (A)]



*It is never about **end result**  
It's always about the **Journey***

**THANKYOU**

#future **IITians**